

Abstract/Paper

Template For Abstracts For The 31st Annual Conference on Geology of Long Island and Metropolitan New York.

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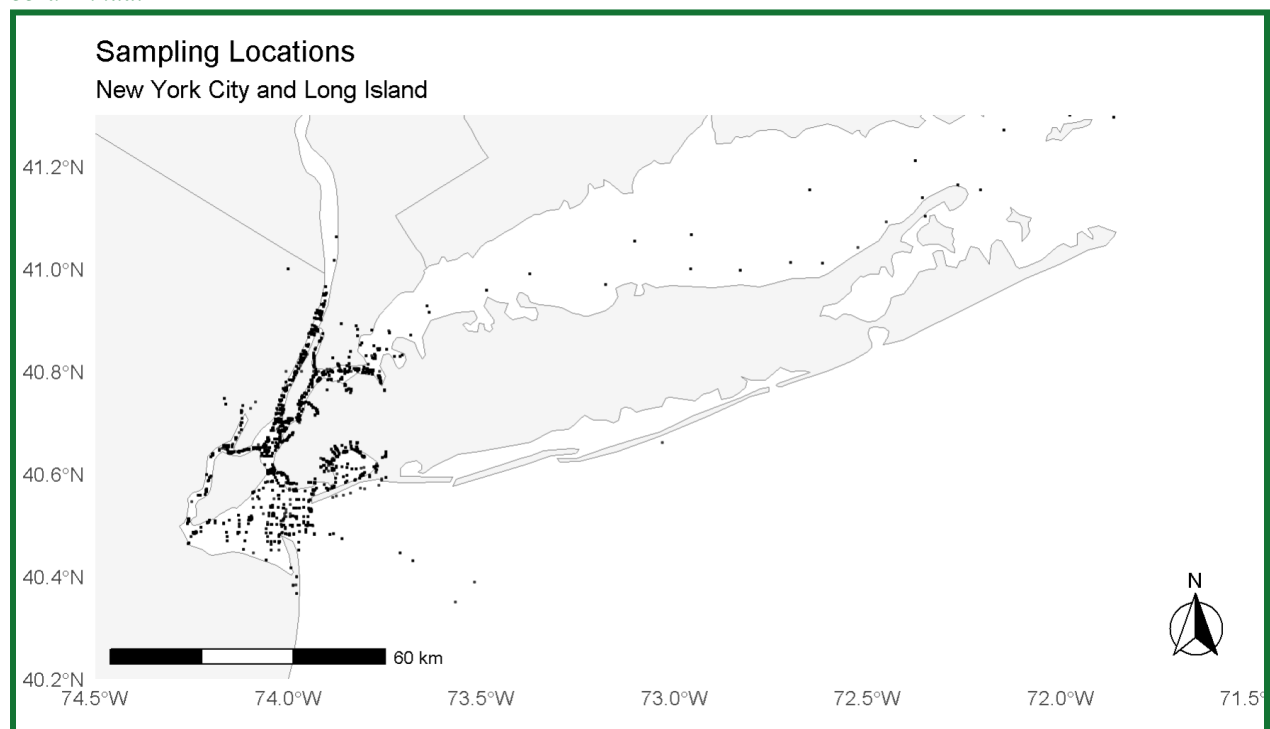
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Title: Impacts of Major Storm Flooding on Harbour Water Quality in New York City using GIS and Time Series Analysis.

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Conference on the Geology of Long Island and Metropolitan New York, April 2026

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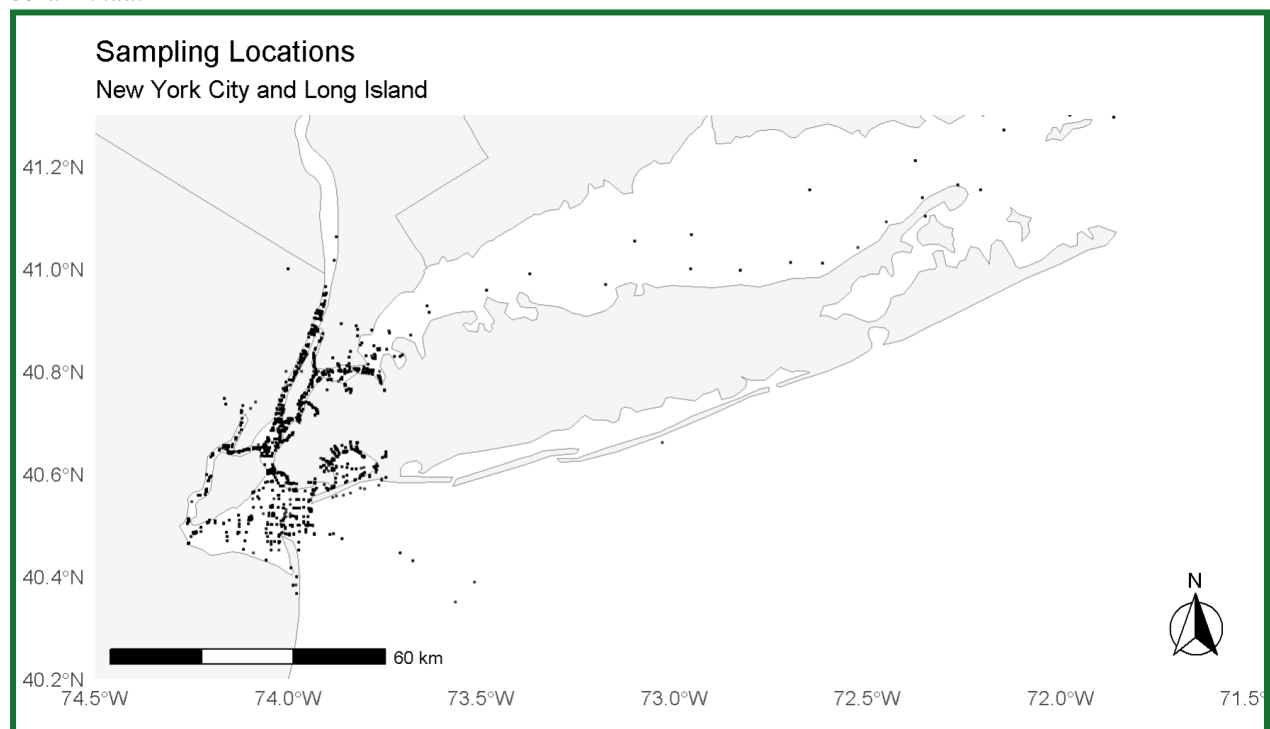
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Introduction: Major storm events in New York City frequently produce intense flooding, especially in low-lying and highly urbanized areas. These floods can overwhelm combined sewer systems and increase pollutants in surrounding waterways, potentially degrading urban water quality. This study investigates whether major flood-producing storms lead to measurable increases in water quality contaminants at monitoring sites, particularly those located near flood-prone zones in NYC.

Methodology:

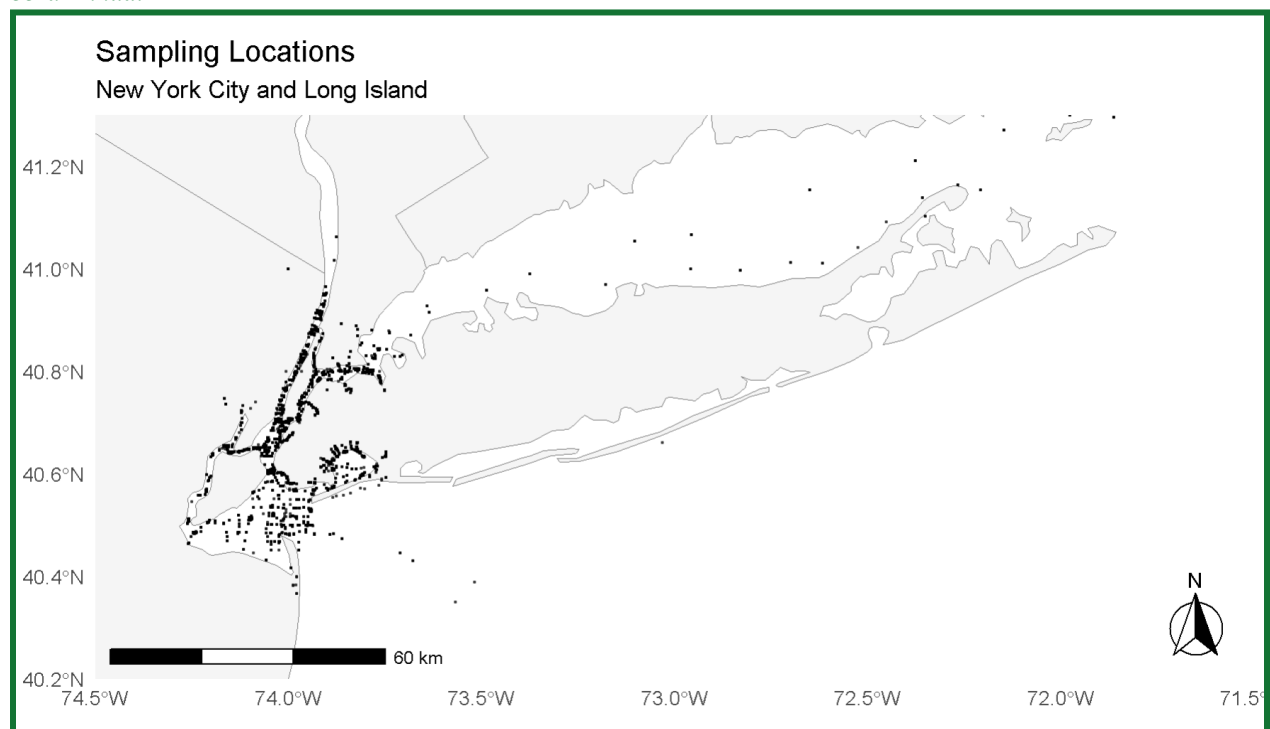
Water quality data was obtained from the NYC Department of Environmental Protection (DEP) Harbor Water Quality survey, accessed via the NYC Open Data portal (https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data). The dataset spans from 1909 to December 2024 and includes measurements collected at multiple monitoring stations across New York City Harbor. Sampling frequency follows a weekly schedule from May through October and monthly from November through April. Four contaminants were selected for analysis : Enterococcus and Fecal Coliform as direct indicators of sewage contamination and public health risk, Turbidity as a measure of physical water disturbance from sediment and debris, and Nitrate serving as an indicator of fertilizer and



nutrient runoff from land. The four storms that were selected were storms that had direct impact on the greater New York City area. The four selected additionally were selected to represent a range of different flooding types: Tropical Storm Fay (July 10, 2020), Tropical Storm Ophelia (September 29, 2023), Hurricane Sandy (October 29, 2012) and Hurricane Ida (September 1, 2021). These events represent three different storm categories, tropical storms, hurricanes, and catastrophic storm surge. This allowed for comparison across different flood-producing mechanisms.

All data processing and statistical analysis were performed in R (R Core Team, 2024) using the tidyverse (Wickham et al., 2019), ggplot2 and lubridate (Grolemund and Wickham, 2011) packages. Raw contaminant values were converted from character to numeric format prior to analysis. For each storm event, a ± 14 day window was applied around the storm landfall date to capture water quality conditions both before and after the event, while also minimizing the influence of unrelated weather events outside that period. Daily mean contaminant values were calculated by averaging readings across all active monitoring stations for each date, providing a representative harbor-wide water quality conditions. Each storm window was divided into two periods: the 14 days before the storm made landfall and the 14 days after. To determine whether water quality changed significantly between these two periods, a Wilcoxon rank-sum test was applied to each storm and contaminant combination, resulting in 16 total statistical comparisons. This specific test was chosen because water quality measurements are highly variable and do not follow a classic normal distribution, making it a more reliable option than a standard t-test for this type of data. Results were considered statistically significant at $p < 0.05$. To investigate whether contaminant levels continued to change in the days following each storm, a Spearman rank correlation was performed between the number of days elapsed since landfall and enterococcus concentrations for the post-storm period of each event. The Spearman method was used due to the same non-normal distribution considerations as mentioned above.

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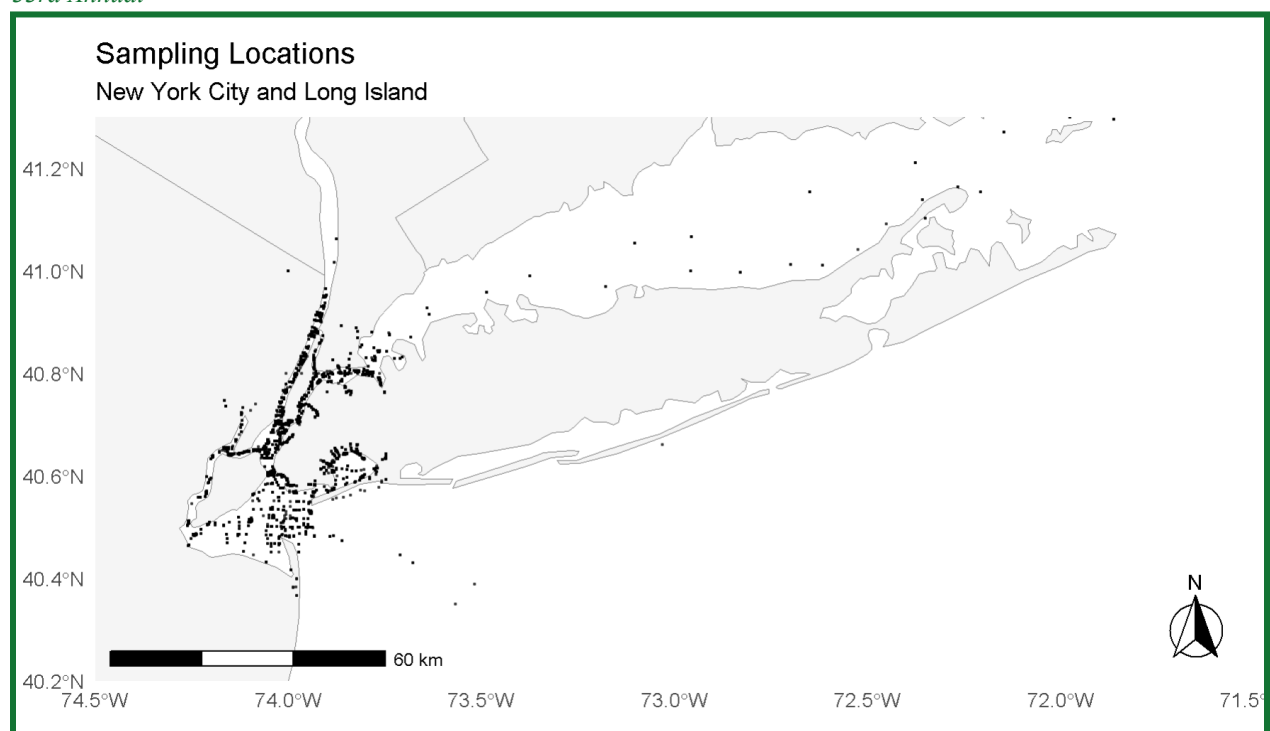
Following initial analysis, an unexpected pattern was observed in which Hurricane Sandy produced bacterial increases after the storm while the three rainfall-dominated storms produced decreases or no significant change. Notably, Hurricane Ida, classified as a Category 4 hurricane, also showed a decrease in bacterial levels following the storm. This suggests that storm intensity alone does not determine the direction of water quality response. Instead, the primary flooding mechanism appears to play a larger role, as Sandy's impacts on NYC were driven largely by storm surge, whereas Ida, Fay, and Ophelia primarily produced heavy rainfall. This led to an additional comparative analysis examining the average percentage change in bacterial concentrations between pre- and post-storm periods, grouped by storm flooding mechanism : Storm surge versus rainfall dominated events.

FEMA National Flood Hazard Layer (NFHL) shapefiles for New York City were obtained from the FEMA Map Service Center (<https://msc.fema.gov>). Monitoring station coordinates were exported from the NYC DEP dataset and will be imported into ArcGIS Pro as point features. A spatial join will be performed to identify which monitoring stations fall within designated FEMA flood zones. This spatial classification will allow for a reanalysis of water quality responses grouped by flood zone designation, providing a more spatially explicit assessment of storm impacts across the harbor. (REFERENCES DOWN BELOW)

Results: N/A

Discussion: This study will help clarify how extreme rainfall and flooding influence urban coastal water quality in a densely developed city with aging sewer infrastructure. Understanding these relationships may highlight the vulnerability of flood-prone neighborhoods and adjacent waterways to contamination following major storms.

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Conclusion: This project integrates time series analysis with GIS-based spatial assessment to link storm events and flood exposure. The findings are expected to contribute to improved understanding of urban flood impacts on coastal environments and may inform future stormwater management and public health protection strategies in New York City.

To all teams: This is a suggested statement please revise with your names and work accordingly

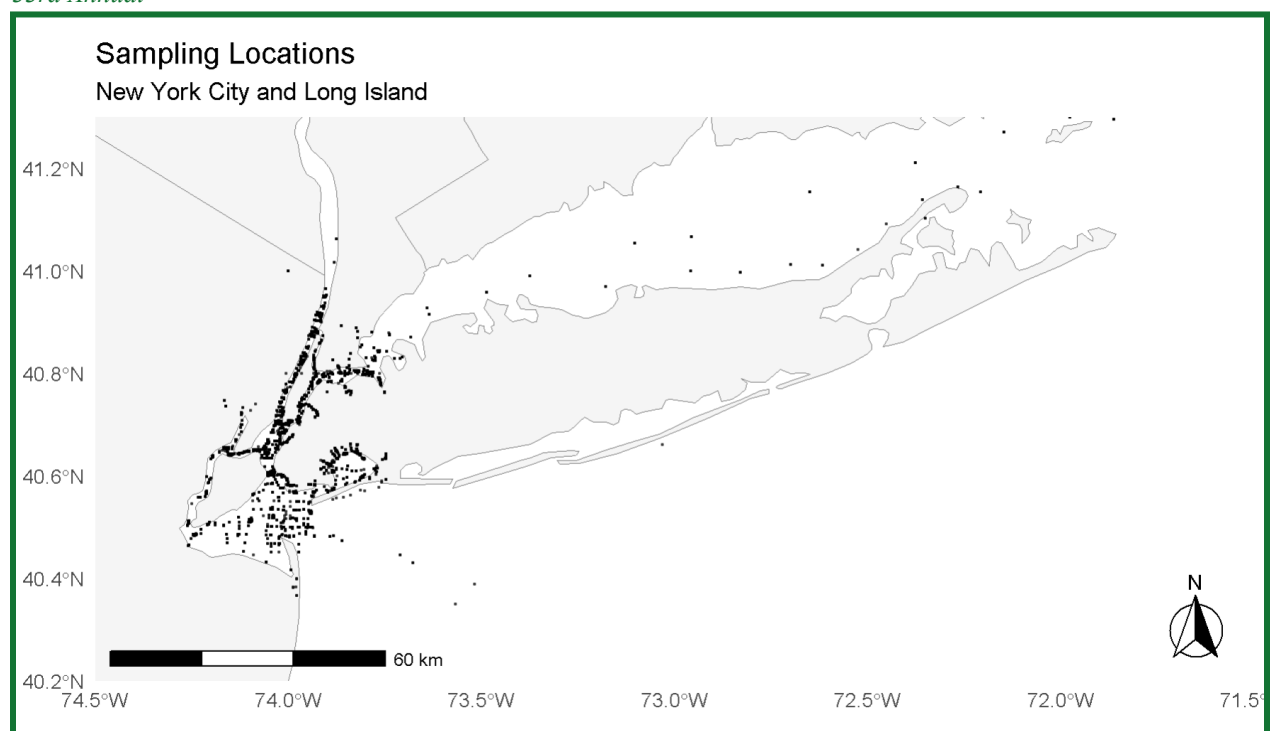
Credit Authorship Contribution Statement

Badger, T.: editing, literature, Rstudio coding, Figure 4, Figure 5, writing- Results, Discussion; Parag, A.: Literature, editing, writing- Global Mapper, Google Earth, Abstract, Results- figure 1, Methods, Discussion; Pelletier, A.: references, literature, editing, writing - Introduction, Discussion; DeRocchis, S.: figures, editing, USGS Graphs, Global Mapper, writing - Methods, Discussion; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing; Tsakiri, K.G.: Rstudio coding.

References:

R Core Team (2024). R : A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org/> Weicham, H., Averick, M., Bryan, J., Chang, W., McGowan, L.D., Francois, R., Grolemond G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T.L., Miller, E., Bache, S.M., Muller, K., ooms, J., Robinson, D.,

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Seidel, D.P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.
<https://joss.theoj.org/papers/10.21105/joss.01686> Wickham, H. (2019). *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>

Data water quality:

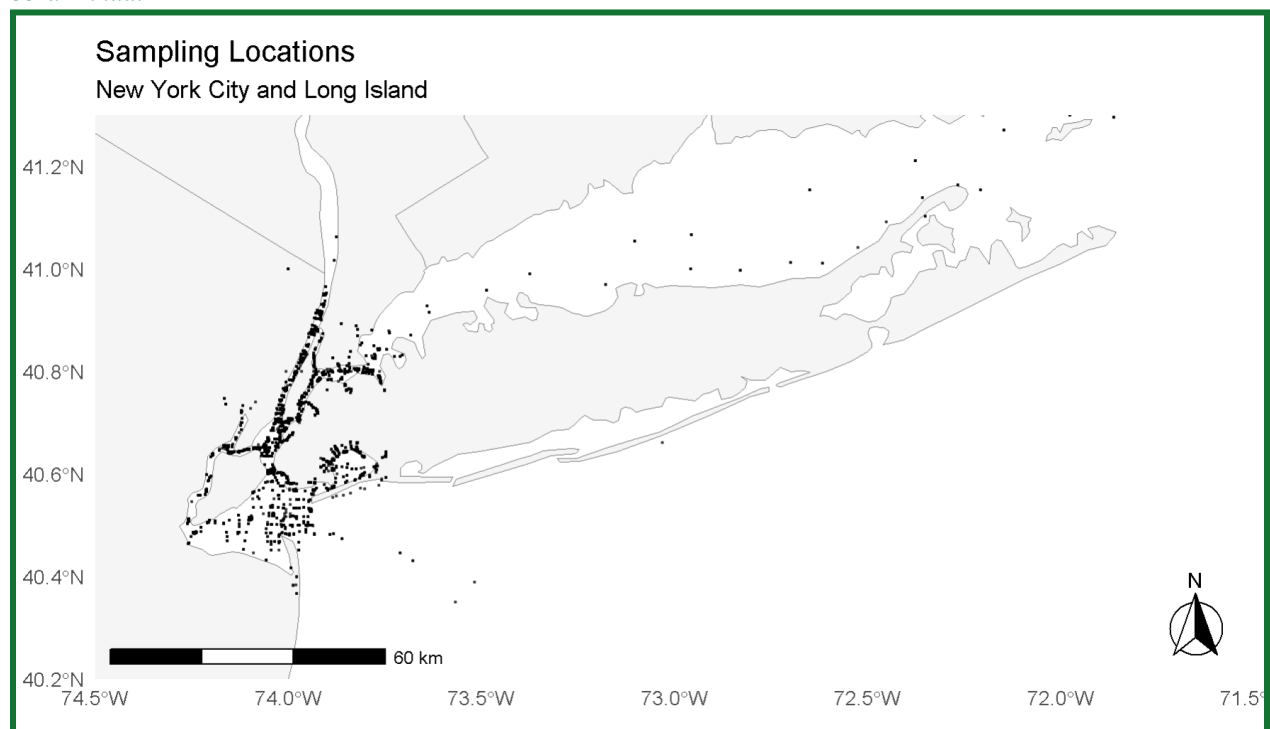
https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data

For arcgis fema data:

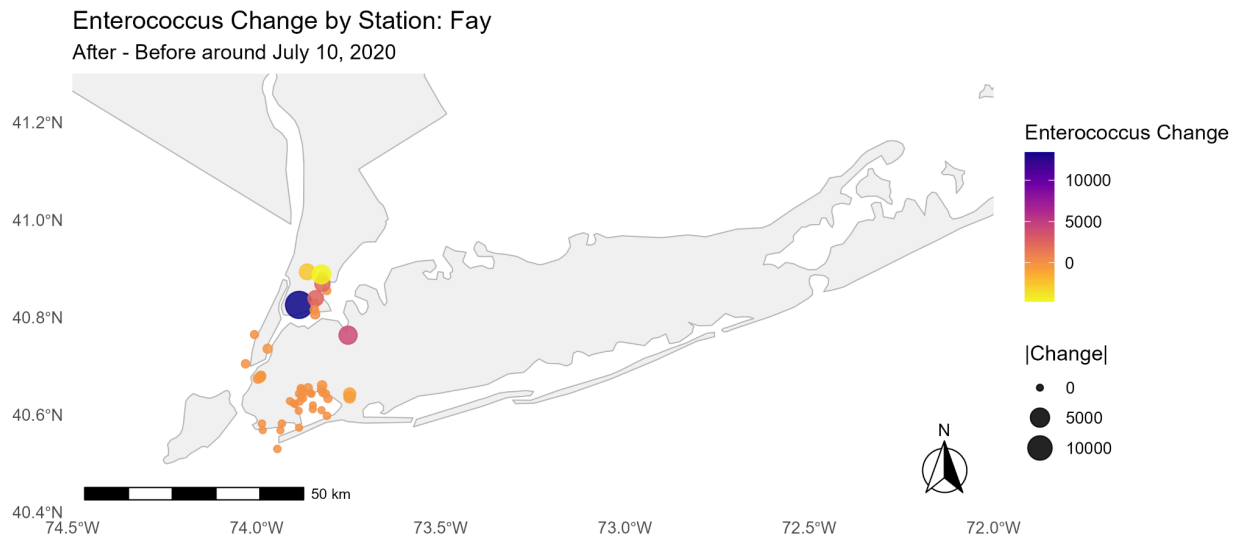
<https://msc.fema.gov/portal/search?AddressQuery=nyc>

Dr.M: Some useful figures:

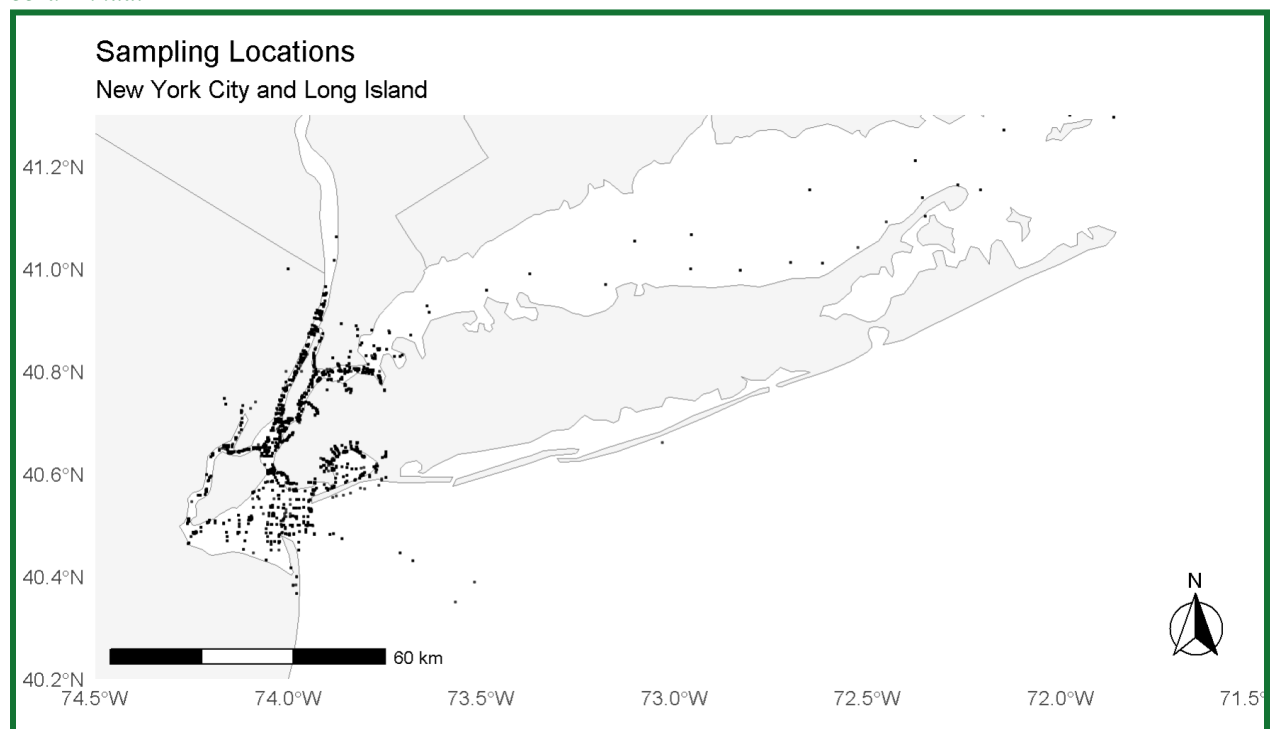
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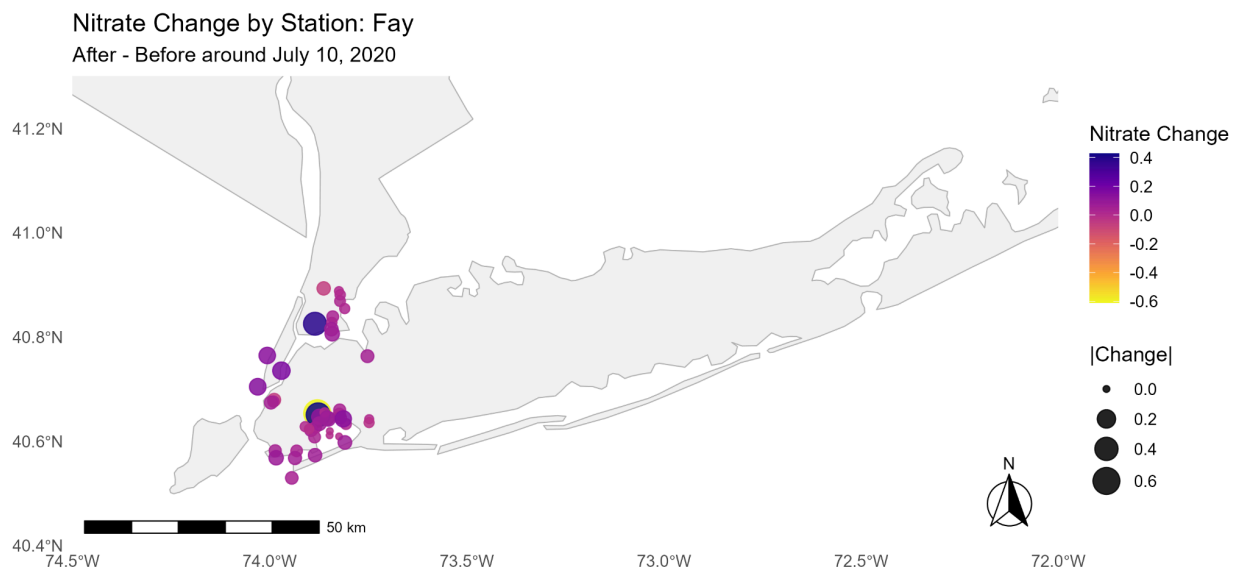


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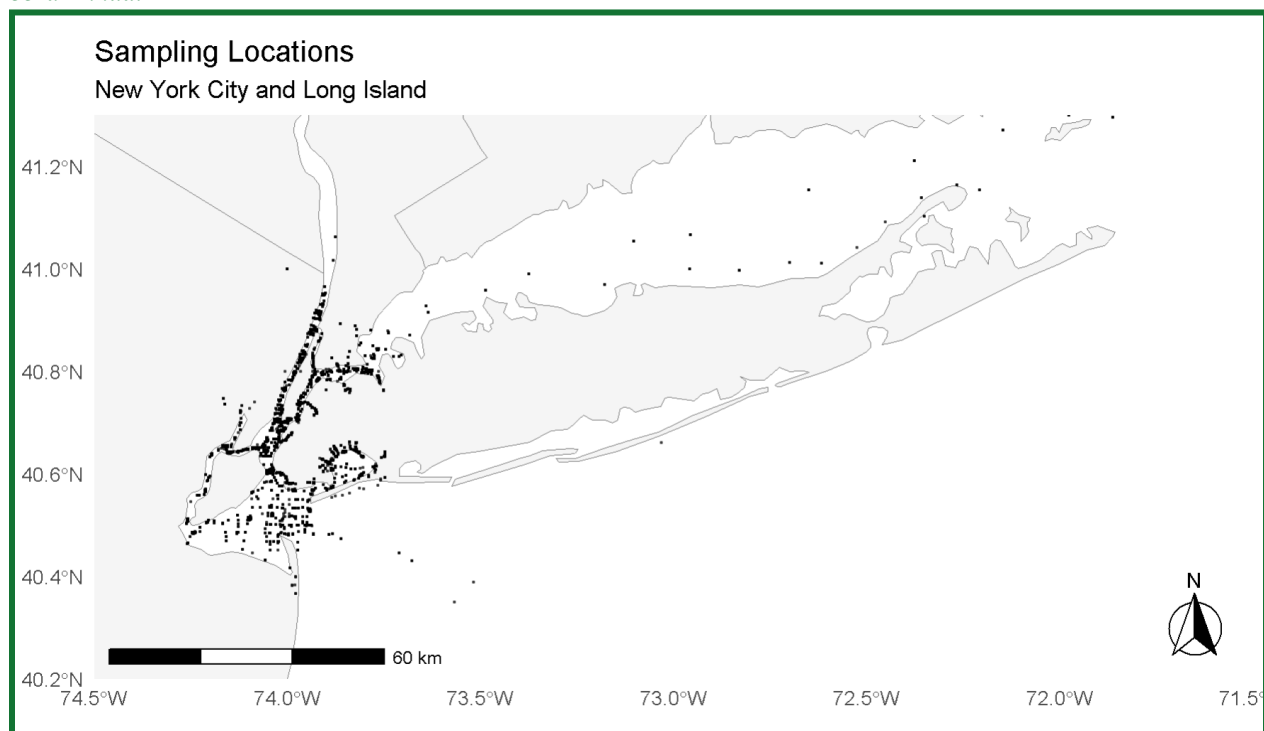


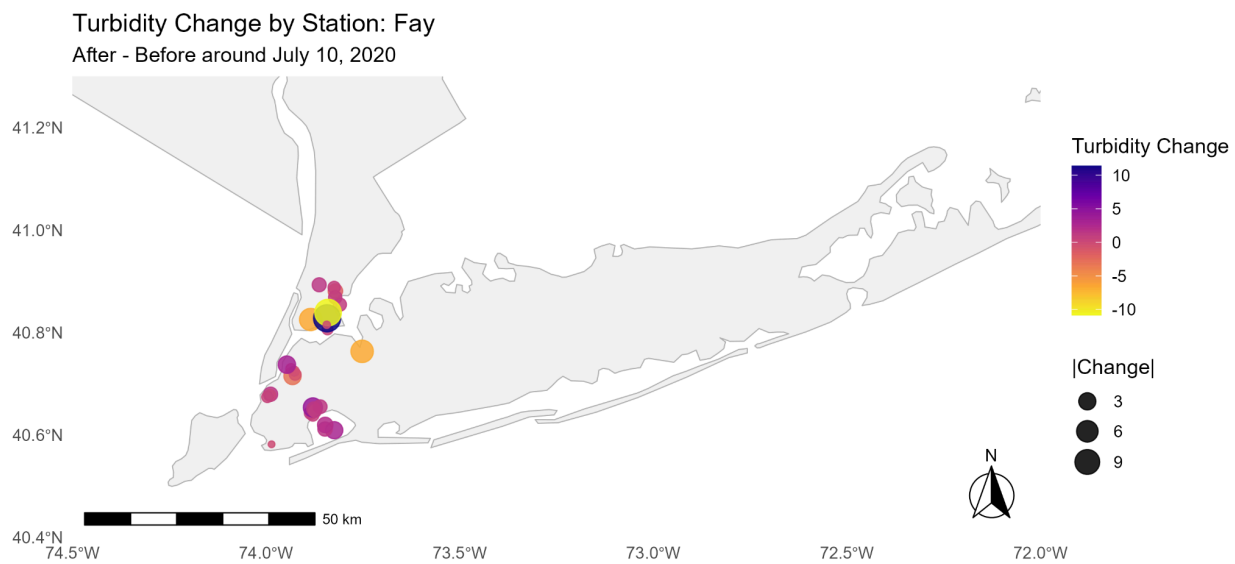
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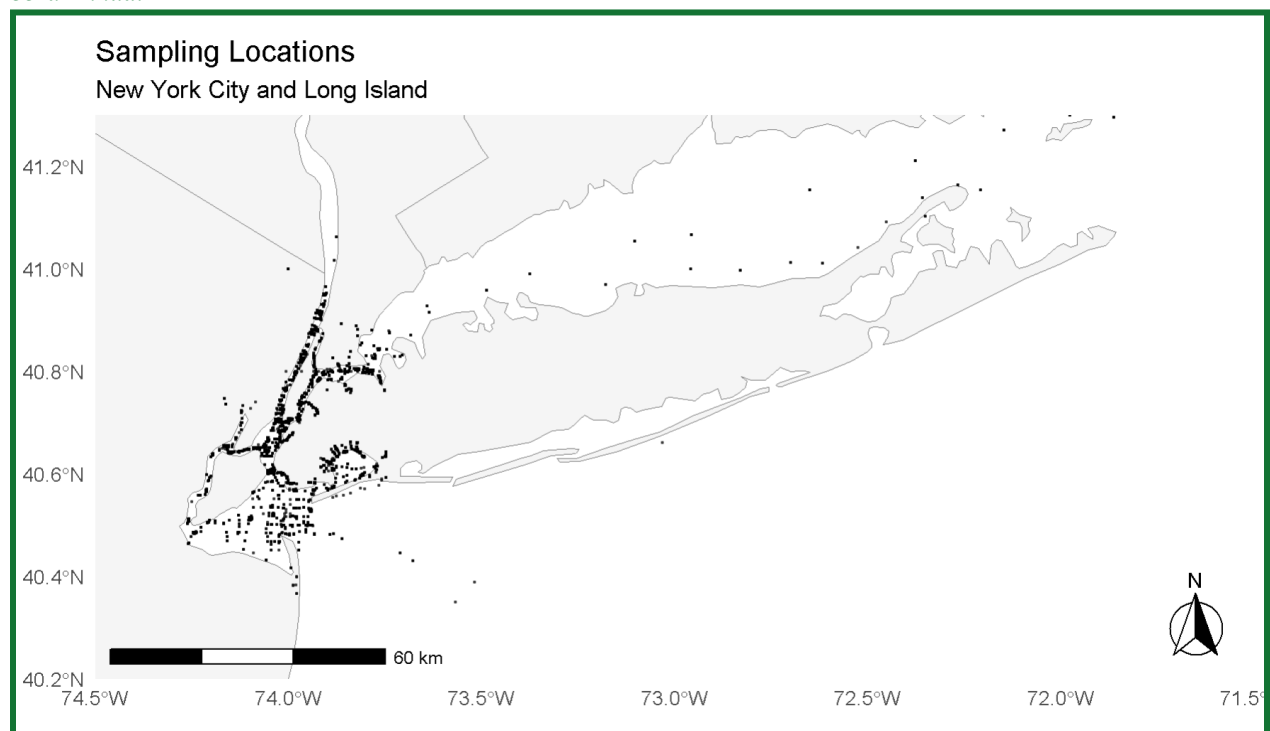


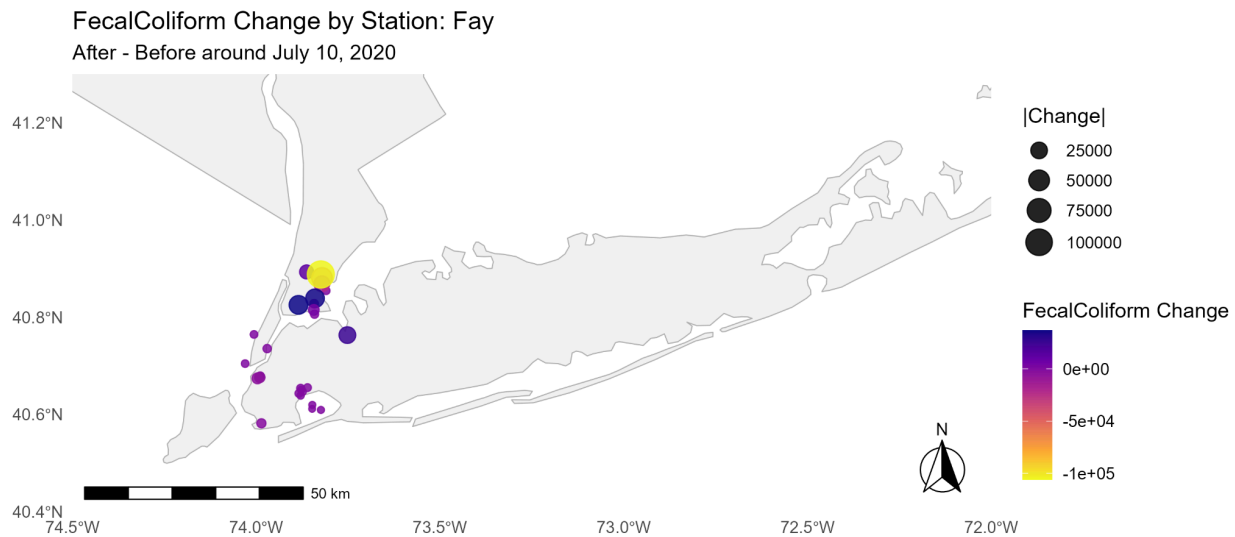
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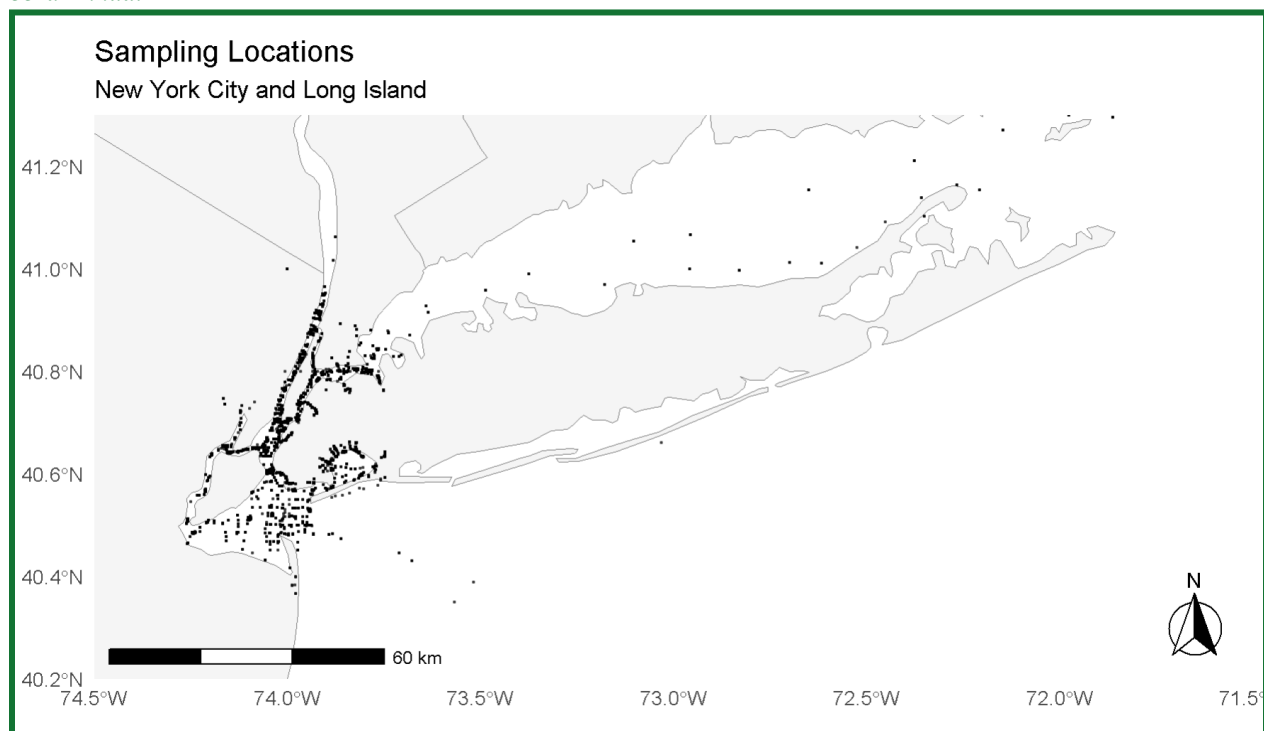


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Wilcoxon Signed-Rank test

Dr.M here is some info that you will need:

1 Why does the Wilcoxon test use the median instead of the mean?

The short answer:

Because the Wilcoxon test is **rank-based** and does not assume normality — it tests for differences in central tendency without relying on the mean.

Let's unpack that.

◆ **Mean-based tests (like t-tests)**

The **t-test**:

- Compares **means**
- Assumes **normal distribution**
- Is sensitive to **outliers**
- Relies on actual numeric distances between values

The mean works well when:

- Data are symmetric
 - No extreme outliers
 - Approximate normality
-

◆ **Wilcoxon tests (rank-based)**

Wilcoxon tests:

- Convert values into **ranks**
- Use the **ordering**, not the raw magnitudes
- Do **not assume normality**
- Are much less sensitive to outliers

Because they rely on ranks rather than actual distances:

👉 They test for differences in **location**

👉 In symmetric distributions, this corresponds to differences in the **median**

♦ Why not the mean?

The mean depends on:

- Exact distances between numbers
- Distribution shape

But Wilcoxon ignores distances and uses ranks instead.

Since ranks reflect **relative position**, not magnitude, the test naturally aligns with the **median (central position)**, not the mean.

♦ Important nuance

Wilcoxon does **not directly test “median difference”** in the strict mathematical sense.

It tests:

Whether one distribution tends to produce larger values than the other.

If distributions are similarly shaped and symmetric, this becomes equivalent to testing for a **difference in medians**.

2 Are Wilcoxon and Mann–Whitney exactly the same test?

Short answer:

Almost — but technically different in origin.

◆ Mann–Whitney U test

- Developed by Mann & Whitney (1947)
 - Based on the **U statistic**
 - Used for **two independent samples**
-

◆ Wilcoxon Rank-Sum test

- Developed by Wilcoxon (1945)
 - Also for **two independent samples**
 - Based on the **sum of ranks**
-

◆ Key point

For two independent samples:

Wilcoxon Rank-Sum test = Mann–Whitney U test

They are mathematically equivalent — just expressed with different statistics.

U statistic and rank-sum statistic are directly convertible:

$$U = W - n_1(n_1 + 1)/2$$

Where:

- W = Wilcoxon rank sum
- n_1 = sample size of group 1

Same p-value. Same conclusion.

But don't confuse with:

♦ Wilcoxon Signed-Rank Test

This is different.

Used for:

- Paired samples
- One-sample median test
- Before/after measurements

So:

Test	Samples	Equivalent To
Wilcoxon Rank-Sum	Independent	Mann–Whitney U
Mann–Whitney U	Independent	Wilcoxon Rank-Sum
Wilcoxon Signed-Rank	Paired	Not the same

1 Why is it called “signed”?

Because it uses the **sign of the differences**.

Step 1: Compute paired differences

For each subject:

$$d_i = \text{After}_i - \text{Before}_i \quad d_i = \text{After}_i - \text{Before}_i$$

Each difference has:

- A **magnitude** (how big the change is)
 - A **sign** (+ or -)
-

Step 2: Remove zeros

If $d_i = 0$, that pair is removed.

Step 3: Rank the absolute differences

We rank:

$$|d_i| \quad |d_j| \quad |d_k|$$

Smallest absolute difference → rank 1

Next → rank 2

etc.

Step 4: Reattach the sign

Now we give each rank the original sign of its difference.

So we get:

- Positive ranks (if $\text{After} > \text{Before}$)

- Negative ranks (if After < Before)

Step 5: Compare sum of positive and negative ranks

The test statistic is based on:

$W^+ = \text{sum of positive ranks}$
 $W^- = \text{sum of negative ranks}$

If there is **no real change**, positive and negative ranks should be similar.

If there is a real increase or decrease, one side will dominate.

♦ Why "signed"?

Because unlike the simple sign test, this test:

- Uses **sign**
- AND uses **rank magnitude**

So it's a **signed + rank** test.

2 What is the hypothesis?

The Wilcoxon signed-rank test evaluates whether the **median of the differences is zero**.

Null hypothesis H_0

Median of $d_i = 0$

Meaning:

There is no systematic change between before and after.

Alternative hypothesis H_A

Three possibilities:

Two-sided:

Median of $d_i \neq 0$ \text{Median of } d_i \neq 0

There is a change (increase or decrease).

One-sided (increase):

Median of $d_i > 0$ \text{Median of } d_i > 0

After values tend to be larger.

One-sided (decrease):

Median of $d_i < 0$ \text{Median of } d_i < 0

After values tend to be smaller.

3 How to interpret the p-value?

The p-value tells you:

If the true median difference were zero, how likely is it to observe this imbalance in signed ranks?

♦ If p-value < 0.05

Reject H_0

Conclusion:

There is statistically significant evidence of a change between before and after.

If median difference $> 0 \rightarrow$ increase

If median difference $< 0 \rightarrow$ decrease

◆ If p-value > 0.05

Fail to reject H_0

Conclusion:

There is not enough evidence to conclude a change.

Important:

This does NOT prove there is no effect.

It means we don't have strong enough evidence.

4 What is it really testing conceptually?

It tests whether:

- Positive and negative differences are balanced.
- The distribution of differences is symmetric around zero.

If symmetric and centered at zero $\rightarrow$ no change.

5 Why not use paired t-test instead?

Paired t-test:

- Tests mean difference

- Assumes normality of differences

Wilcoxon signed-rank:

- Tests median difference
- Does not require normal distribution
- More robust to outliers

Important Assumption

The Wilcoxon signed-rank test assumes:

The distribution of differences is symmetric around the median.

If differences are very skewed, interpretation becomes more complicated.

Concept	Meaning
t	
Signed	Uses direction (+/−) of differences
Rank	Uses magnitude via ranking
H_0	Median difference = 0
$p < 0.05$	Evidence of change
$p > 0.05$	No strong evidence of change

Here is an example for your research:

Small illustrative results table (example)

Below is a **made-up** small table showing how you would read results for 3 variables (this is an illustrative example — *not* from the real dataset). Each row is one site × variable using 14 paired

days (after aggregated daily medians); zeros should be removed meaning no conclusive result about median difference.

Variable	n_pai rs	Median_befo re	Median_aft er	Median_diff (after-befor e)	#positi ve diffs	#negati ve diffs	Su m pos ran ks (W ⁺)	Su m neg ran ks (W ⁻)	W (statisti c)	p-v alue	Interpretatio n (α=0.05)
Enterococc us	12	45	110	+65	9	3	85	28	28 (or 85)	0.0 04	Significant increase — after values tend to be larger; reject H ₀ .
Turbidity	14	4.2	6.0	+1.8	8	6	60	46	46	0.1 2	Not significant at 0.05 — weak evidence of change.
Nitrate	10	0.15	0.12	-0.03	3	7	22	48	22	0.0 2	Significant decrease — after values tend to be smaller; reject H ₀ .

Talk2me-Dr.M

Dr.M-Mar11th please see the other tabs for more information (Wilcoxon..)

Concerns

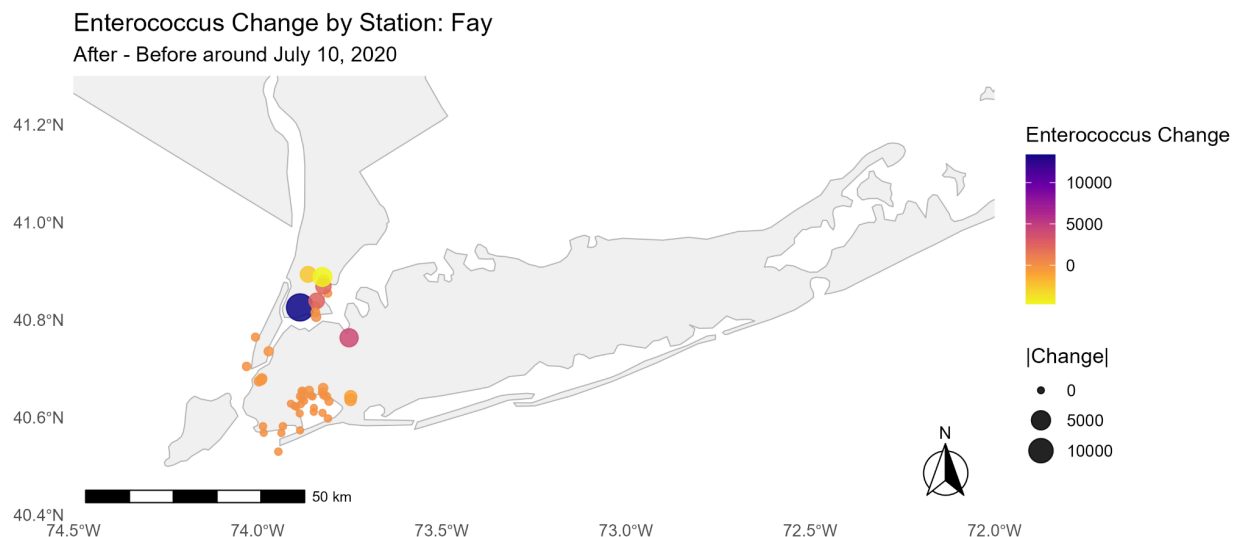
1. Samples located outside New York or in Upstate are considered in your analysis?

This is acceptable for a general trend, but it removes spatial variability (important limitation).

2. You are averaging all data (daily) from all sites → missing the extreme values.

Just to keep something straight: Wilcoxon should be applied to **raw observations**, not daily means—averaging reduces variability and sample size. You have visualize averages on your time series plots, but test the raw data → mismatch between what's shown and what's tested. Testing is correct but visualizing averages does confuse - just remember this.

3. Your first map shows multiple locations while the GIS maps show localized locations close to NYC. I guess there were no analysis at those locations (you need to double check). See below:



abstract final

Impacts of Major Storm Flooding on Harbor Water Quality in New York City using Time Series and Spatial Analysis

Carvan, S.C.¹, Carvan, Z.C¹, and Marsellos, A.E.¹

¹*Department of Geology, Environment and Sustainability, Hofstra University*

Abstract:

Water quality degradation following major storm events poses significant public health risks in urbanized coastal environments. This study examines the impacts of four storm events on bacterial and chemical contaminant levels in New York City Harbor using water quality data obtained from the NYC Department of Environmental Protection Harbor Water Quality Survey (1909-2024). The four storms analyzed were Hurricane Sandy (2012), Tropical Storm Fay (2020), Hurricane Ida (2021), and Tropical Storm Ophelia (2023), selected to represent a range of storm types and flooding mechanisms. For each event, a ± 14 -day window was applied around the storm landfall date and Wilcoxon rank-sum tests were used to compare pre- and post-storm concentrations of Enterococcus, Fecal Coliform, Turbidity, and Nitrate across 16 statistical comparisons. Station-level spatial maps were produced using Inverse Distance Weighted interpolation in R to visualize the geographic distribution of contaminant change across the harbor. Results indicate that storm flooding mechanism, rather than storm intensity alone, is a primary driver of water quality response. Hurricane Sandy, a surge-dominated event, produced significant post-storm bacterial increases of 814% in Enterococcus and 236% in Fecal Coliform, with Spearman correlation confirming contamination continued rising for days after landfall ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage infrastructure damage. In contrast, rainfall-dominated storms including Hurricane Ida, Tropical Storm Fay, and Tropical Storm Ophelia produced dilution effects or no significant change, despite Ida being a Category 4 hurricane. These findings suggest that flood mechanism is a stronger predictor of harbor water quality response than storm intensity, with direct implications for post-storm public health monitoring and stormwater management in coastal urban systems.

Introduction:

Major storm events in New York City frequently produce intense flooding, especially in low-lying and highly urbanized areas. These floods can overwhelm combined sewer systems and increase pollutants in surrounding waterways, potentially degrading urban water quality. New York City's aging sewer infrastructure is particularly vulnerable to storm surge events, which push ocean water inland and can force raw or partially treated sewage directly into the harbor. This study investigates whether major flood-producing storms lead to measurable changes in water quality contaminants at monitoring sites across NYC Harbor, and whether the type of flooding mechanism, storm surge versus rainfall, determines the direction of that water quality response.

Methodology:

Water quality data were obtained from the NYC Department of Environmental Protection (DEP) Harbor Water Quality Survey, accessed via the NYC Open Data portal (https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data). The dataset spans from 1909 to December 2024 and includes measurements collected at multiple monitoring stations across New York City Harbor. Sampling frequency follows a weekly schedule from May through October and monthly from November through April. Four contaminants were selected for analysis: Enterococcus and Fecal Coliform as direct indicators of sewage contamination and public health risk, Turbidity as a measure of physical water disturbance from sediment and debris, and Nitrate as an indicator of fertilizer and nutrient runoff from land.

Four storms that made direct impact on the greater New York City area were selected to represent a range of flooding types: Tropical Storm Fay (July 10, 2020), Tropical Storm Ophelia (September 29, 2023), Hurricane Ida (September 1, 2021), and Hurricane Sandy (October 29, 2012). These events represent three storm categories: tropical storms, hurricane, and catastrophic storm surge, allowing for comparison across different flood-producing mechanisms.

All data processing and statistical analysis were performed in R (R Core Team, 2024) using the tidyverse (Wickham et al., 2019), ggplot2 (Wickham, 2016), and lubridate (Grolemund and Wickham, 2011) packages. Raw contaminant values were converted from character to numeric format prior to analysis. For each storm event, a ± 14 -day window was applied around the storm landfall date to capture water quality conditions both before and after the event, while minimizing the influence of unrelated weather events outside that period. Contaminant observations were aggregated by sampling date, averaging readings across all active monitoring stations, providing a harbor-wide representation of water quality conditions.

Each storm window was divided into two periods: the 14 days before the storm made landfall and the 14 days after. To determine whether water quality changed significantly between these two periods, a Wilcoxon rank-sum test was applied to each storm and contaminant combination, resulting in 16 total statistical comparisons. This test was chosen because water quality measurements are highly variable and do not follow a normal distribution, making it a more reliable option than a standard t-test for this type of data. Results were considered statistically significant at $p < 0.05$.

To investigate whether contaminant levels continued to change in the days following each storm, a Spearman rank correlation was performed between the number of days elapsed since landfall and Enterococcus concentrations for the post-storm period of each event. The Spearman method was used due to the same non-normal distribution considerations as above.

Following initial analysis, an unexpected pattern was observed in which Hurricane Sandy produced bacterial increases after the storm while the three rainfall-dominated storms produced decreases or no significant change. Notably, Hurricane Ida, classified as a Category 4 hurricane, also showed a decrease in bacterial levels following the storm. This suggests that storm intensity alone does not determine the direction of the water quality response. Instead, the primary flooding mechanism appears to play a larger role, as Sandy's impacts on NYC were driven largely by storm surge, whereas Ida, Fay, and Ophelia primarily produced heavy rainfall. This led to an additional comparative analysis examining the average percentage change in bacterial concentrations between pre- and post-storm periods, grouped by storm flooding mechanism: storm surge versus rainfall-dominated events.

Spatial analysis was conducted using R (R Core Team, 2024) with the `sf`, `gstat`, `sp`, `viridis`, `rnatrualearth`, and `ggspatial` packages. Coordinates in the dataset were cleaned and corrected for any swapped latitude and longitude values prior to mapping. Station-level before and after change maps were produced for each storm and contaminant, with Inverse Distance Weighted (IDW) interpolation applied to visualize the spatial distribution of contaminant change across the harbor. Federal Emergency Management Agency (FEMA) National Flood Hazard Layer (NFHL) shapefiles for New York City were obtained from the FEMA Map Service Center (<https://msc.fema.gov>). Monitoring station coordinates were exported from the NYC Department of Environmental Protection (DEP) Harbor Water Quality Survey, available via the NYC Open Data Portal (https://data.cityofnewyork.us/Environment/Harbor-Water-Quality/5uug-f49n/about_data), as point features. A planned spatial join between monitoring stations and FEMA flood zone designations will allow for a future reanalysis of water quality responses grouped by flood zone, providing a more spatially precise assessment of storm impacts across the harbor.

Inverse Distance Weighted (IDW) interpolation was selected due to its simplicity and suitability for datasets with irregularly spaced monitoring stations and moderate spatial density. Unlike geostatistical methods such as kriging, IDW does not require assumptions about spatial autocorrelation structure or variogram modeling, making it appropriate for exploratory spatial analysis where the primary goal is visualization of localized patterns rather than formal spatial prediction. Given the limited number of monitoring stations and uneven spatial distribution across NYC Harbor, IDW provides an interpretable method for representing contaminant gradients.

Results:

Time series plots display contaminant values aggregated by sampling date across all monitoring stations for visualization purposes, while Wilcoxon rank-sum tests were applied to raw individual station observations.

Wilcoxon rank-sum tests comparing pre- and post-storm contaminant concentrations yielded statistically significant results for at least two contaminants in each of the four storm events across 16 total comparisons (Figure 1).

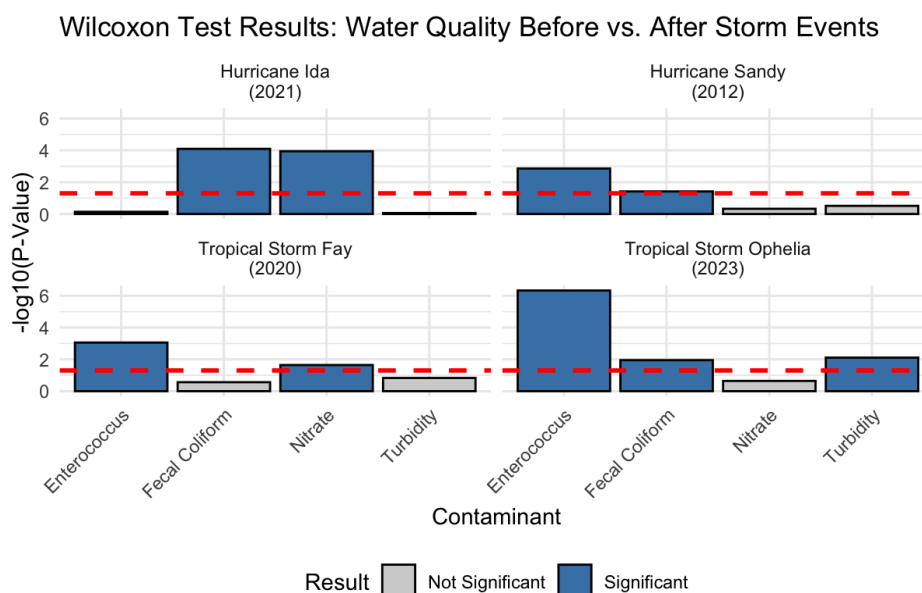


Figure 1. Wilcoxon rank-sum test results for four water quality contaminants across four storm events. Bars represent the negative log10 transformed p-value for each contaminant and storm combination. The red dashed line indicates the significance threshold at $p = 0.05$. Blue bars indicate statistically significant results and gray bars indicate non-significant results. P-values ranged from 4.71×10^{-7} (Ophelia, Enterococcus) to 0.87 (Ida, Turbidity), with 8 of 16 comparisons reaching statistical significance..

For Hurricane Sandy, significant post-storm increases were observed in Enterococcus ($p = 0.0014$) and Fecal Coliform ($p = 0.038$), while Turbidity ($p = 0.31$) and Nitrate ($p = 0.47$) did not reach significance. Average Enterococcus concentrations increased by 814% and Fecal Coliform by 236% in the 14 days following landfall. Spearman correlation confirmed that Enterococcus levels continued to rise with each passing day after Sandy ($\rho = 0.62$, $p = 0.0003$), indicating prolonged rather than immediate contamination.

For Tropical Storm Fay, significant changes were observed in Enterococcus ($p = 0.0009$) and Nitrate ($p = 0.023$), while Fecal Coliform ($p = 0.27$) and Turbidity ($p = 0.15$) did not reach significance. Spearman correlation showed a strong positive trend in Enterococcus in the days following the storm ($\rho = 0.86$, p less than 0.0001), though the overall magnitude of change was substantially smaller than Sandy.

For Hurricane Ida, significant changes were found in Fecal Coliform ($p = 0.00008$) and Nitrate ($p = 0.0001$), while Enterococcus ($p = 0.73$) and Turbidity ($p = 0.87$) showed no significant change. Unlike Sandy, average bacterial concentrations decreased following Ida, with Fecal Coliform declining by 43% and Enterococcus showing no meaningful directional trend (Spearman $\rho = -0.04$, $p = 0.64$).

For Tropical Storm Ophelia, significant changes were observed in Enterococcus ($p = 0.0000005$), Fecal Coliform ($p = 0.011$), and Turbidity ($p = 0.0078$), while Nitrate ($p = 0.23$) did not reach significance. Ophelia produced the strongest dilution response of all four storms, with Enterococcus declining by 87% and Fecal Coliform by 69% following landfall. Spearman correlation confirmed a significant declining trend in Enterococcus in the days after the storm ($\rho = -0.29$, $p = 0.008$).

Comparative analysis of average bacterial percentage change grouped by flooding mechanism revealed a clear contrast between storm types (Figure 2, surge vs. rainfall chart). Hurricane Sandy, the only surge-dominated event, produced large post-storm bacterial increases across both indicators. The three rainfall-dominated storms, Tropical Storm Fay, Hurricane Ida, and Tropical Storm Ophelia, produced either decreases or negligible changes in bacterial concentrations, consistent with a freshwater dilution effect. This pattern held regardless of storm intensity, as Hurricane Ida, a Category 4 hurricane, produced bacterial decreases comparable to the weaker tropical storms.

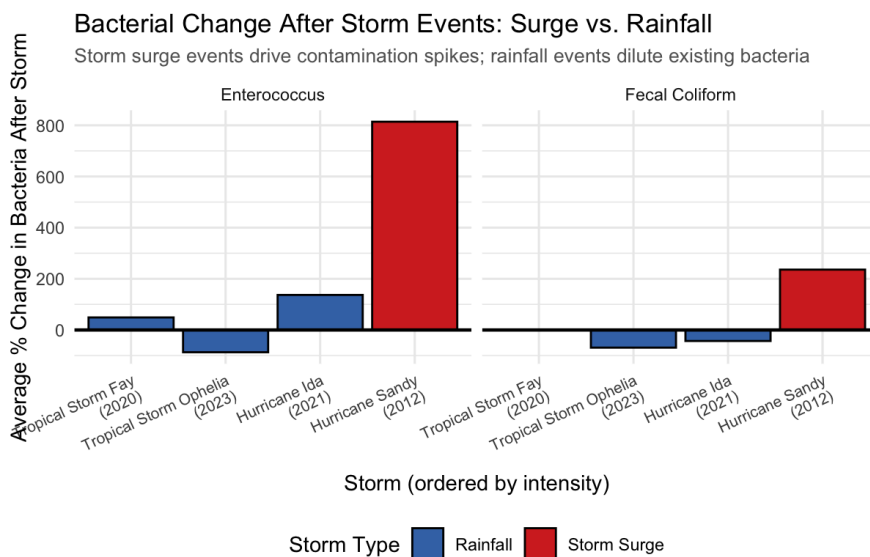


Figure 2. Average percentage change in Enterococcus and Fecal Coliform concentrations in the 14 days following each storm event compared to the 14 days prior, grouped by storm flooding mechanism. Red bars indicate the surge-dominated Hurricane Sandy and blue bars indicate rainfall-dominated storms. Storms are ordered left to right by increasing intensity

Station-level spatial maps produced using IDW interpolation revealed variability in contaminant response across individual monitoring stations within the harbor (Figure 3, Fay Spatial Map). These maps indicate that the harbor-wide averaged results may mask localized contamination hotspots at specific stations, particularly those in closer proximity to flood-prone areas.

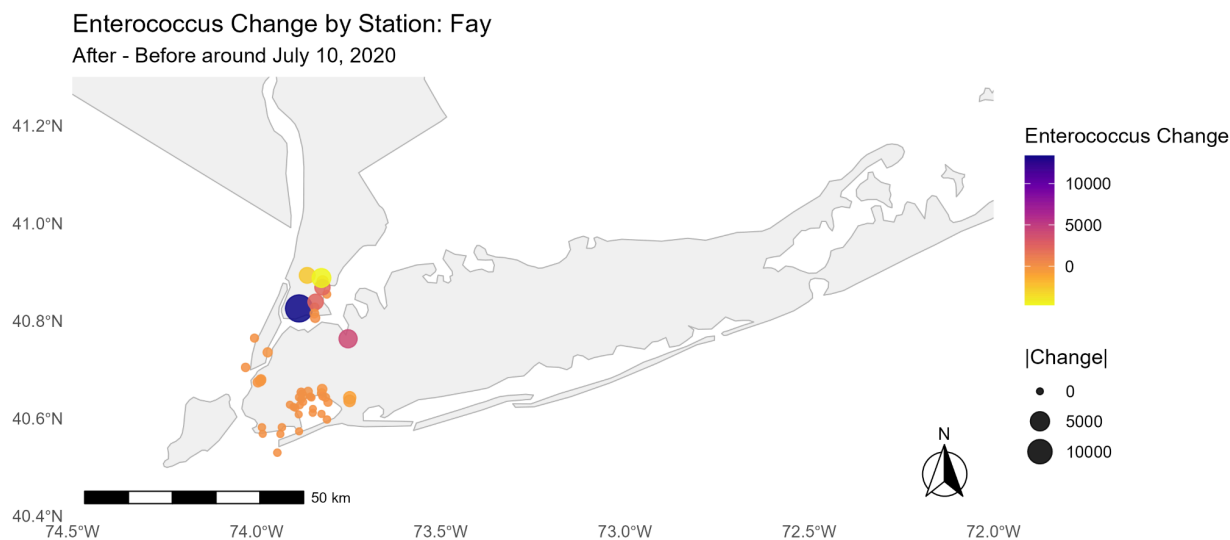


Figure 3. Station-level *Enterococcus* change map for Tropical Storm Fay (July 10, 2020), produced using Inverse Distance Weighted (IDW) interpolation in R. Each point represents an individual NYC DEP monitoring station, with color indicating the direction and magnitude of change in *Enterococcus* concentrations (cells/100mL) between pre- and post-storm periods with warmer colors indicating increases and cooler colors indicating decreases. Dot size reflects the absolute magnitude of change. Spatial maps for remaining storm events are currently being finalized.

Discussion:

This study found that storm flooding mechanism, rather than storm intensity, was the primary driver of water quality change in NYC Harbor. Hurricane Sandy, the only surge-driven storm in this analysis, produced the largest bacterial increases, with *Enterococcus* rising 814% and Fecal Coliform rising 236% in the 14 days following landfall. Spearman correlation confirmed these levels continued rising after the storm ($\rho = 0.62$, $p = 0.0003$), suggesting ongoing sewage system damage rather than a single-day event. In contrast, all three rainfall-dominated storms showed bacterial dilution or no significant change, including Hurricane Ida despite being a Category 4 hurricane, which supports the conclusion that storm category alone is not a reliable predictor of water quality outcomes.

Several limitations should be noted. Daily averages were used for visualization while Wilcoxon tests were applied to raw observations, which creates a mismatch between what is shown and what is tested. Averaging across all stations also removes spatial variability and may mask localized contamination near flood-prone areas. Future analysis incorporating FEMA flood zone designations at the individual station level would strengthen these findings considerably.

Conclusion:

Storm surge events pose a significantly greater risk to harbor water quality than rainfall-dominated storms, regardless of storm intensity. These findings have practical implications for post-storm public health monitoring in coastal cities with aging sewer infrastructure. Future work will integrate FEMA flood zone data with station-level water quality responses to provide a more spatially precise assessment of storm impacts across New York City Harbor.

Credit Authorship Contribution Statement

Carvan, S.: Data collection, statistical analysis, RStudio coding, writing of Abstract, Introduction, Methods, Results, Discussion, Conclusion, and Figures 1, 2, 3; editing.

Carvan, Z.: Data collection, statistical analysis, writing of Methods, Results, and Discussion; data analysis, editing, and contribution to Figure 2.

Marsellos, A.E.: Project supervision, RStudio coding, spatial analysis, review and revision of manuscript, and contribution to Figure 3.

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Tab 6

Poster:

Abstract:

Water quality degradation following major storm events poses significant public health risks in urbanized coastal environments. This study examines the impacts of four storm events on bacterial and chemical contaminant levels in New York City Harbor using water quality data from the NYC Department of Environmental Protection Harbor Water Quality Survey (1909–2024). The storms analyzed—Hurricane Sandy (2012), Tropical Storm Fay (2020), Hurricane Ida (2021), and Tropical Storm Ophelia (2023)—represent different storm types and flooding mechanisms. A ±14-day window around each landfall was used with Wilcoxon rank-sum tests to compare pre- and post-storm concentrations of Enterococcus, Fecal Coliform, Turbidity, and Nitrate. Station-level spatial maps were created using Inverse Distance Weighted interpolation to visualize contaminant distribution across the harbor. Storm flooding mechanism, not intensity alone, is the primary driver of water quality response. Hurricane Sandy, a surge-dominated event, produced significant post-storm bacterial increases (814% Enterococcus, 236% Fecal Coliform), with Spearman correlation showing contamination rising for days after landfall ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage infrastructure damage. Rainfall-dominated storms—Hurricane Ida, Tropical Storm Fay, and Tropical Storm Ophelia—produced dilution effects or no significant change, despite Ida being Category 4. These findings demonstrate that flood mechanism is a stronger predictor of harbor water quality response than storm intensity, with implications for post-storm public health monitoring and coastal stormwater management.

Methods

NYC DEP Harbor Water Quality Survey (1909–2024)
-Weekly/monthly sampling across monitoring stations
-Contaminants: Enterococcus, Fecal Coliform, Turbidity, Nitrate

* Storm Events
-Hurricane Sandy (2012, surge-dominated)
-Tropical Storm Fay (2020, rainfall-dominated)
-Hurricane Ida (2021, rainfall-dominated, Category 4)
-Tropical Storm Ophelia (2023, rainfall-dominated)

* RStudio Analysis
±14-day window around storm landfall
-Wilcoxon rank-sum tests ($p < 0.05$)
-Spearman rank correlation (temporal trends)
-Percentage change by flooding mechanism

* Spatial Mapping
-Inverse Distance Weighted interpolation
-Station-level before/after contaminant maps

NYC DEP Harbor Water Quality Survey (1909–2024)
Weekly/monthly sampling across monitoring stations
Contaminants: Enterococcus, Fecal Coliform, Turbidity, Nitrate
Storm Events
Hurricane Sandy (2012, surge-dominated)
Tropical Storm Fay (2020, rainfall-dominated)
Hurricane Ida (2021, rainfall-dominated, Category 4)
Tropical Storm Ophelia (2023, rainfall-dominated)
RStudio Analysis
±14-day window around storm landfall
Wilcoxon rank-sum tests ($p < 0.05$)
Spearman rank correlation (temporal trends)
Percentage change by flooding mechanism
Spatial Mapping
Inverse Distance Weighted interpolation
Station-level before/after contaminant maps

Results

Wilcoxon rank-sum tests: 8 of 16 comparisons significant ($p < 0.05$)
Hurricane Sandy (Surge-Dominated)
Enterococcus: +814% ($p = 0.0014$)
Fecal Coliform: +236% ($p = 0.038$)
Spearman correlation showed prolonged contamination days after landfall ($\rho = 0.62$, $p = 0.0003$)

Rainfall-Dominated Storms (Fay, Ida, Ophelia)
Fay: Enterococcus increased ($p = 0.0009$)
Ida: Fecal Coliform decreased 43% ($p = 0.00008$) despite Category 4 intensity
Ophelia: Enterococcus decreased 87% ($p = 0.0000005$), Fecal Coliform decreased 69% ($p = 0.011$)

Key Finding
Flooding mechanism (surge vs. rainfall) is the primary driver of water quality response, not storm intensity
Station-level spatial maps revealed localized contamination hotspots across the harbor

Background

New York City Harbor receives runoff from a heavily urbanized watershed with aging combined sewer infrastructure, making it vulnerable to contamination during storm events. Hurricane Sandy resulted in 11 billion gallons of untreated and partially treated sewage entering regional waterways due to record storm surge overwhelming treatment facilities (Climate Central, 2013). As climate change intensifies both storm surge and rainfall events, understanding their distinct impacts on coastal water quality is increasingly critical. Previous studies have focused on storm intensity as a predictor of water quality degradation, but the role of flooding mechanism, whether surge-driven or rainfall-dominated, remains poorly understood.

Conclusion/discussion

Storm flooding mechanism, not storm intensity, was the primary driver of water quality change in NYC Harbor
Hurricane Sandy (surge-driven) produced the largest bacterial increases: Enterococcus +814%, Fecal Coliform +236%
Spearman correlation confirmed contamination continued rising after Sandy ($\rho = 0.62$, $p = 0.0003$), suggesting prolonged sewage system damage
All three rainfall-dominated storms showed bacterial dilution or no significant change, including Category 4 Hurricane Ida
Averaging across all stations removes spatial variability and may mask localized contamination near flood-prone areas
Future analysis will incorporate FEMA flood zone designations at the individual station level for a more spatially precise assessment

Acknowledgment

The authors acknowledge Dr. Antonios E. Marsellos and his Geoinformatics Laboratory at Hofstra University for fostering an interdisciplinary research environment that integrates geoinformatics, artificial intelligence, and geohazard analysis, enabling the successful completion of this study.

Authorship

Carvan, S.: Data collection, statistical analysis, RStudio coding, writing of Abstract, Introduction, Methods, Results, Discussion, Conclusion, and Figures 1, 2, 3; editing.
Carvan, Z.: Data collection, statistical analysis, writing of Methods, Results, and Discussion; data analysis, editing, and contribution to Figure 2.
Marsellos, A.E.: Project supervision, RStudio coding, spatial analysis, review and revision of manuscript, and contribution to Figure 3.

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Figure 1

Wilcoxon rank-sum test results showing statistical significance of water quality changes across four storms. Blue bars indicate significant results ($p < 0.05$); gray bars indicate non-significant results. 8 of 16 comparisons reached significance.

Figure 3

Station-level Enterococcus change map for Tropical Storm Fay using Inverse Distance Weighted interpolation. Warmer colors indicate increases; cooler colors indicate decreases. Dot size reflects magnitude of change.

Figure

Average percentage change in Enterococcus and Fecal Coliform concentrations post-storm compared to pre-storm, grouped by flooding mechanism. Red bars indicate surge-dominated Hurricane Sandy; blue bars indicate rainfall-dominated storms.

Sampling Locations: NYC DEP monitoring stations across New York City Harbor and surrounding waters. Black dots indicate station locations used in water quality analysis ($n = 16$ stations).